

Explanation of Each Category

1. Fully Observable / Partially Observable:

- **Fully Observable:** The agent has complete information about the environment at any time.
- **Partially Observable:** The agent has incomplete information about the environment.

2. Deterministic / Stochastic:

- **Deterministic:** The outcome of an action is predictable and consistent.
- **Stochastic:** The outcome of an action is probabilistic and may vary.

3. Episodic / Sequential:

- **Episodic:** The agent's experience consists of a series of discrete, independent episodes.
- **Sequential:** The agent's experience is a continuous sequence of actions and states, where each action can affect future states.

4. Static / Dynamic:

- **Static:** The environment does not change while the agent is making a decision.
- **Dynamic:** The environment changes continuously, requiring the agent to adapt.

5. Discrete / Continuous:

- **Discrete:** The agent has a finite number of possible actions and states.
- **Continuous:** The agent has an infinite number of possible actions and states.

6. Single Agent / Multiple Agents:

- **Single Agent:** The agent operates alone without interaction with other agents.
- **Multiple Agents:** The agent operates in an environment with other agents, requiring coordination and interaction.